

9 Reset and clock control (RCC)

9.1 Reset

There are three types of reset, defined as system reset, power reset and RTC domain reset.

9.1.1 Power reset

A power reset is generated when one of the following events occurs:

1. Power-on/power-down reset (POR/PDR reset)
2. When exiting Standby mode

A power reset sets all registers to their reset values except the RTC domain (see [Figure 8](#)).

9.1.2 System reset

A system reset sets all registers to their reset values unless specified otherwise in the register description.

A system reset is generated when one of the following events occurs:

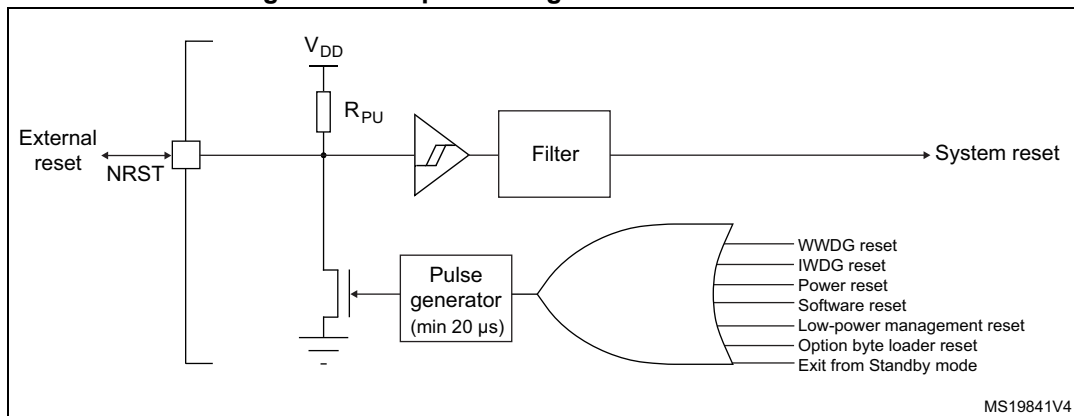
1. A low level on the NRST pin (external reset)
2. Window watchdog event (WWDG reset)
3. Independent watchdog event (IWDG reset)
4. A software reset (SW reset) (see [Software reset](#))
5. Low-power management reset (see [Low-power management reset](#))
6. Option byte loader reset (see [Option byte loader reset](#))
7. A power reset

The reset source can be identified by checking the reset flags in the Control/Status register, RCC_CSR (see [Section 9.4.10: Control/status register \(RCC_CSR\)](#)).

These sources act on the NRST pin and it is always kept low during the delay phase. The RESET service routine vector is fixed at address 0x0000_0004 in the memory map.

The system reset signal provided to the device is output on the NRST pin. The pulse generator guarantees a minimum reset pulse duration of 20 μ s for each internal reset source. In case of an external reset, the reset pulse is generated while the NRST pin is asserted low.

Figure 12. Simplified diagram of the reset circuit



Software reset

The SYSRESETREQ bit in Cortex-M4[®]F application interrupt and reset control register must be set to force a software reset on the device. Refer to the *STM32 Cortex[®]-M4 MCUs and MPUs programming manual* (PM0214) for more details.

Low-power management reset

There are two ways to generate a low-power management reset:

1. Reset generated when entering Standby mode:
This type of reset is enabled by resetting nRST_STDBY bit in User Option Bytes. In this case, whenever a Standby mode entry sequence is successfully executed, the device is reset instead of entering Standby mode.
2. Reset when entering Stop mode:
This type of reset is enabled by resetting nRST_STOP bit in User Option Bytes. In this case, whenever a Stop mode entry sequence is successfully executed, the device is reset instead of entering Stop mode.

For further information on the User Option Bytes, refer to [Section 5: Option byte description](#).

Option byte loader reset

The option byte loader reset is generated when the OBL_LAUNCH bit (bit 13) is set in the FLASH_CR register. This bit is used to launch the option byte loading by software.

9.1.3 RTC domain reset

The RTC domain has two specific resets that affect only the RTC domain ([Figure 8](#)).

An RTC domain reset only affects the LSE oscillator, the RTC, the Backup registers, and the RCC [RTC domain control register \(RCC_BDCR\)](#). It is generated when one of the following events occurs.

1. Software reset, triggered by setting the BDRST bit in the [RTC domain control register \(RCC_BDCR\)](#).
2. V_{DD} power-up if V_{BAT} has been disconnected when it was low.

The backup registers are also reset when one of the following events occurs:

1. RTC tamper detection event.
2. Change of the read out protection from level 1 to level 0.

9.2 Clocks

Three different clock sources can be used to drive the system clock (SYSCLK):

- HSI 8 MHz RC oscillator clock
- HSE oscillator clock
- PLL clock

The devices have the following additional clock sources:

- 40 kHz low speed internal RC (LSI RC) which drives the independent watchdog and optionally the RTC used for Auto-wakeup from Stop/Standby mode.
- 32.768 kHz low speed external crystal (LSE crystal) which optionally drives the real-time clock (RTCCLK)

Each clock source can be switched on or off independently when it is not used, to optimize power consumption.

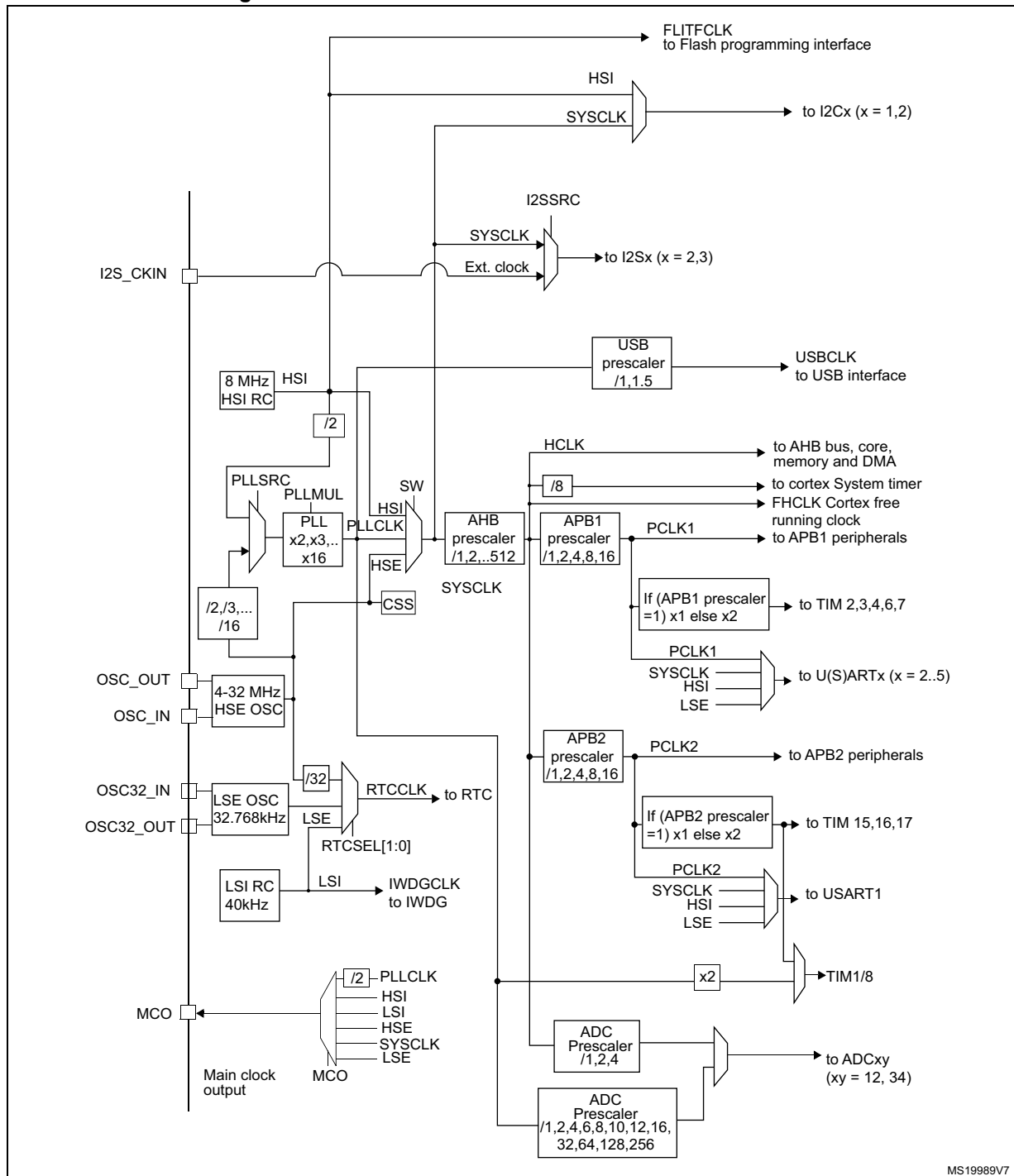
Several prescalers can be used to configure the AHB frequency, the high speed APB (APB2) and the low speed APB (APB1) domains. The maximum frequency of the AHB and APB2 domains is 72 MHz. The maximum allowed frequency of the APB1 domain is 36 MHz.

All the peripheral clocks are derived from their bus clock (HCLK, PCLK1 or PCLK2) except:

- The flash memory programming interface clock (FLITFCLK) which is always the HSI clock.
- The 48-MHz USB clock which is derived from the PLL VCO (STM32F303xB/C/D/E devices)
- The option byte loader clock, which is always the HSI clock
- The ADCs clock, which is derived from the PLL output. It can reach 72 MHz and can then be divided by 1,2,4,6,8,10,12,16,32,64,128 or 256.
- The U(S)ARTs clock, which is derived (selected by software) from one of the four following sources:
 - system clock
 - HSI clock
 - LSE clock
 - APB1 or APB2 clock (PCLK1 or PCLK2 depending on which APB is mapped the USART)
- The I2C1/2 (I2C1/2/3 in STM32F303xD/E and STM32F398xE) clock, which is derived (selected by software) from one of the two following sources:
 - system clock
 - HSI clock
- The I2S2 and I2S3 clocks, which can be derived from an external dedicated clock source.
- The RTC clock which is derived from the LSE, LSI or from the HSE clock divided by 32.
- The IWDG clock, which is always the LSI clock.

The RCC feeds the Cortex System Timer (SysTick) external clock with the AHB clock (HCLK) divided by 8. The SysTick can work either with this clock or directly with the Cortex clock (HCLK), configurable in the SysTick Control and Status Register.

Figure 13. STM32F303xB/C and STM32F358xC clock tree



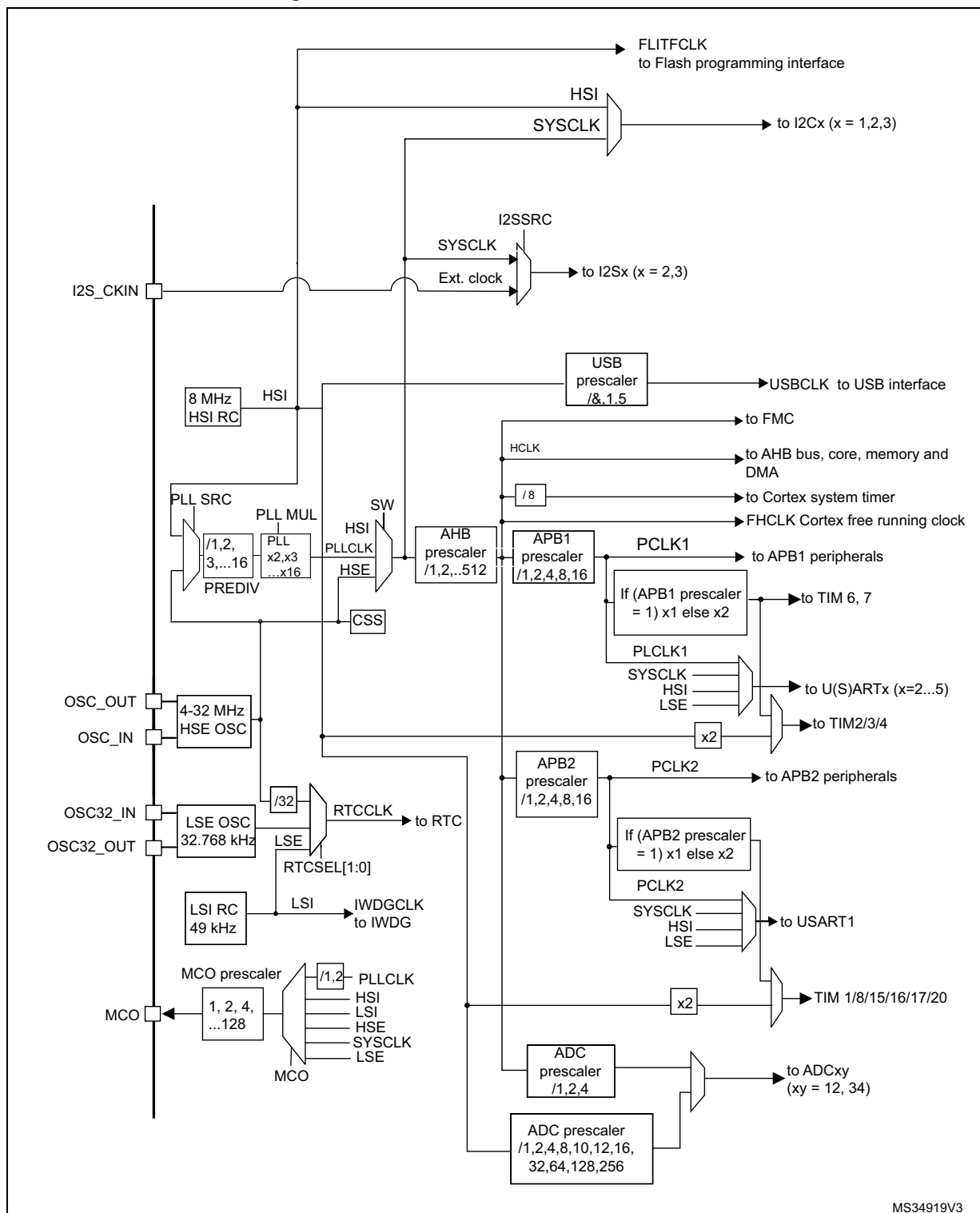
MS19989V7

1. For full details about the internal and external clock source characteristics, refer to the "Electrical characteristics" section in your device datasheet.
2. TIM1 and TIM8 can be clocked from the PLLCLKx2 running up to 144 MHz when the system clock source is the PLL and

the AHB and APB2 prescalers are set to '1'.

- The ADC clock can be derived from the AHB clock of the ADC bus interface, divided by a programmable factor (1, 2 or 4). When the programmable factor is '1', the AHB prescaler must be equal to '1'.

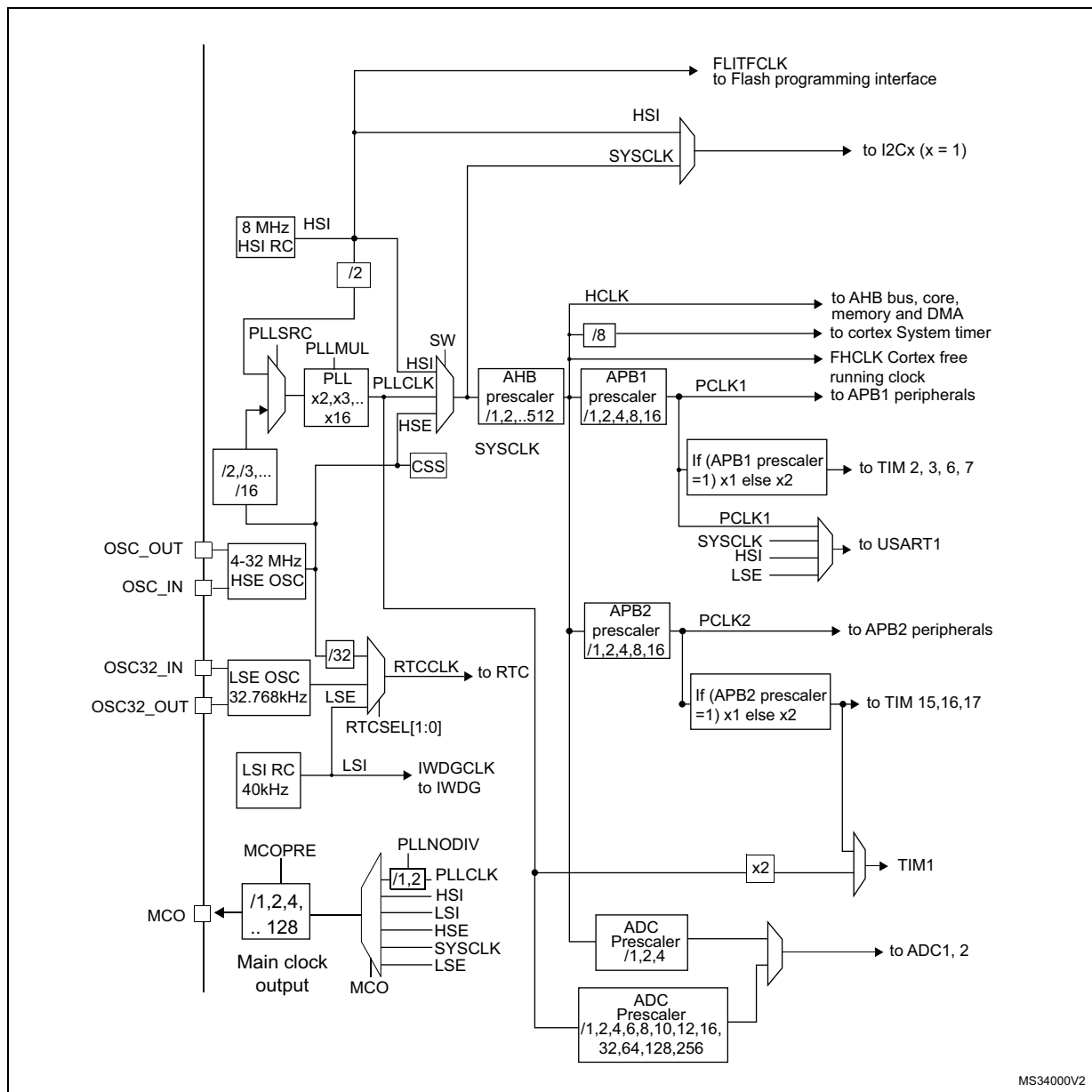
Figure 14. STM32F303xDxE and STM32F398xE clock tree



MS34919V3

1. For full details about the internal and external clock source characteristics, please refer to the Electrical characteristics section in the device datasheet.
2. TIMx (x = 1/2/3/4/8/15/16/17/20) can be clocked from the PLL running at 144 MHz when the system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.
3. The ADC clock can be derived from the AHB clock of the ADC bus interface, divided by a programmable factor (1, 2 or 4). When the programmable factor is "1", the AHB prescaler must be equal to "1".

Figure 15. STM32F303x6/8 and STM32F328x8 clock tree



1. For full details about the internal and external clock source characteristics, refer to the "Electrical characteristics" section in your device datasheet.
2. TIM1 can be clocked from the PLL running at 144 MHz when the system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.
3. The ADC clock can be derived from the AHB clock of the ADC bus interface, divided by a programmable factor (1, 2 or 4). When the programmable factor is '1', the AHB prescaler must be equal to '1'.

FCLK acts as Cortex-M4®F free-running clock. For more details, refer to the *STM32 Cortex®-M4 MCUs and MPUs programming manual* (PM0214).

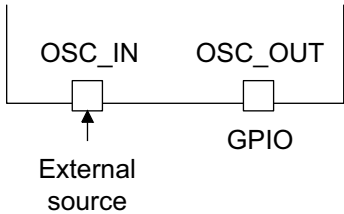
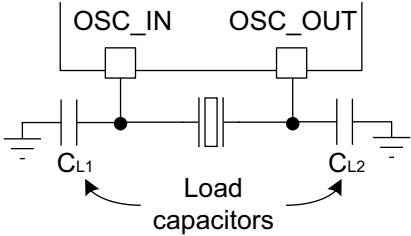
9.2.1 HSE clock

The high speed external clock signal (HSE) can be generated from two possible clock sources:

- HSE external crystal/ceramic resonator
- HSE user external clock

The resonator and the load capacitors have to be placed as close as possible to the oscillator pins to minimize output distortion and startup stabilization time. The loading capacitance values must be adjusted according to the selected oscillator.

Figure 16. HSE/ LSE clock sources

Clock source	Hardware configuration
External clock	 MSv31915V1
Crystal/Ceramic resonators	 MSv31916V1

External crystal/ceramic resonator (HSE crystal)

The 4 to 32 MHz external oscillator has the advantage of producing a very accurate rate on the main clock.

The associated hardware configuration is shown in [Figure 16](#). Refer to the electrical characteristics section of the *datasheet* for more details.

The HSERDY flag in the [Clock control register \(RCC_CR\)](#) indicates if the HSE oscillator is stable or not. At startup, the clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the [Clock interrupt register \(RCC_CIR\)](#).

The HSE Crystal can be switched on and off using the HSEON bit in the [Clock control register \(RCC_CR\)](#).

Caution: To switch ON the HSE oscillator, 512 HSE clock pulses need to be seen by an internal stabilization counter after the HSEON bit is set. Even in the case that no crystal or resonator is connected to the device, excessive external noise on the OSC_IN pin may still lead the oscillator to start. Once the oscillator is started, it needs another 6 HSE clock pulses to complete a switching OFF sequence. If for any reason the oscillations are no more present on the OSC_IN pin, the oscillator cannot be switched OFF, locking the OSC pins from any other use and introducing unwanted power consumption. To avoid such situation, it is strongly recommended to always enable the Clock Security System (CSS) which is able to switch OFF the oscillator even in this case.

External source (HSE bypass)

In this mode, an external clock source must be provided. It can have a frequency of up to 32 MHz. Select this mode by setting the HSEBYP and HSEON bits in the [Clock control register \(RCC_CR\)](#). The external clock signal (square, sinus or triangle) with ~40-60% duty cycle depending on the frequency (refer to the *datasheet*) has to drive the OSC_IN pin while the OSC_OUT pin can be used a GPIO. See [Figure 16](#).

9.2.2 HSI clock

The HSI clock signal is generated from an internal 8 MHz RC Oscillator and can be used directly as a system clock or divided by 2 to be used as PLL input.

The HSI RC oscillator has the advantage of providing a clock source at low cost (no external components). It also has a faster startup time than the HSE crystal oscillator however, even with calibration the frequency is less accurate than an external crystal oscillator or ceramic resonator.

Calibration

RC oscillator frequencies can vary from one chip to another due to manufacturing process variations, this is why each device is factory calibrated by ST for 1% accuracy at $T_A=25^{\circ}\text{C}$.

After reset, the factory calibration value is loaded in the HSICAL[7:0] bits in the [Clock control register \(RCC_CR\)](#).

If the application is subject to voltage or temperature variations this may affect the RC oscillator speed. The user can trim the HSI frequency in the application using the HSITRIM[4:0] bits in the [Clock control register \(RCC_CR\)](#).

For more details on how to measure the HSI frequency variation, refer to [Section 9.2.14: Internal/external clock measurement with TIM16](#).

The HSIRDY flag in the [Clock control register \(RCC_CR\)](#) indicates if the HSI RC is stable or not. At startup, the HSI RC output clock is not released until this bit is set by hardware.

The HSI RC can be switched on and off using the HSION bit in the [Clock control register \(RCC_CR\)](#).

The HSI signal can also be used as a backup source (Auxiliary clock) if the HSE crystal oscillator fails. Refer to [Section 9.2.7: Clock security system \(CSS\) on page 134](#).

9.2.3 PLL

The internal PLL can be used to multiply the HSI or HSE output clock frequency. Refer to [Figure 13](#) and [Clock control register \(RCC_CR\)](#).

The PLL configuration (selection of the input clock, and multiplication factor) must be done before enabling the PLL. Once the PLL is enabled, these parameters cannot be changed.

To modify the PLL configuration, proceed as follows:

1. Disable the PLL by setting PLLON to 0.
2. Wait until PLLRDY is cleared. The PLL is now fully stopped.
3. Change the desired parameter.
4. Enable the PLL again by setting PLLON to 1.

An interrupt can be generated when the PLL is ready, if enabled in the [Clock interrupt register \(RCC_CIR\)](#).

The PLL output frequency must be set in the range 16-72 MHz.

9.2.4 LSE clock

The LSE crystal is a 32.768 kHz Low Speed External crystal or ceramic resonator. It has the advantage of providing a low-power but highly accurate clock source to the real-time clock peripheral (RTC) for clock/calendar or other timing functions.

The LSE crystal is switched on and off using the LSEON bit in [RTC domain control register \(RCC_BDCR\)](#). The crystal oscillator driving strength can be changed at runtime using the LSEDRV[1:0] bits in the [RTC domain control register \(RCC_BDCR\)](#) to obtain the best compromise between robustness and short start-up time on one side and low-power-consumption on the other.

The LSERDY flag in the [RTC domain control register \(RCC_BDCR\)](#) indicates whether the LSE crystal is stable or not. At startup, the LSE crystal output clock signal is not released until this bit is set by hardware. An interrupt can be generated if enabled in the [Clock interrupt register \(RCC_CIR\)](#).

Caution: To switch ON the LSE oscillator, 4096 LSE clock pulses need to be seen by an internal stabilization counter after the LSEON bit is set. Even in the case that no crystal or resonator is connected to the device, excessive external noise on the OSC32_IN pin may still lead the oscillator to start. Once the oscillator is started, it needs another 6 LSE clock pulses to complete a switching OFF sequence. If for any reason the oscillations are no more present on the OSC_IN pin, the oscillator cannot be switched OFF, locking the OSC32 pins from any other use and introducing unwanted power consumption. The only way to recover such situation is to perform the RTC domain reset by software.

External source (LSE bypass)

In this mode, an external clock source must be provided. It can have a frequency of up to 1 MHz. Select this mode by setting the LSEBYP and LSEON bits in the [RTC domain control register \(RCC_BDCR\)](#). The external clock signal (square, sinus, or triangle) with ~50% duty cycle has to drive the OSC32_IN pin while the OSC32_OUT pin can be used as GPIO. See [Figure 16](#).

9.2.5 LSI clock

The LSI RC acts as a low-power clock source that can be kept running in Stop and Standby mode for the independent watchdog (IWDG) and RTC. The clock frequency is around 40 kHz (between 30 kHz and 50 kHz). For more details, refer to the electrical characteristics section of the datasheets.

The LSI RC can be switched on and off using the LSION bit in the [Control/status register \(RCC_CSR\)](#).

The LSIRDY flag in the [Control/status register \(RCC_CSR\)](#) indicates if the LSI oscillator is stable or not. At startup, the clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the [Clock interrupt register \(RCC_CIR\)](#).

9.2.6 System clock (SYSCLK) selection

Three different clock sources can be used to drive the system clock (SYSCLK):

- HSI oscillator
- HSE oscillator
- PLL

After a system reset, the HSI oscillator is selected as system clock. When a clock source is used directly or through the PLL as a system clock, it is not possible to stop it.

A switch from one clock source to another occurs only if the target clock source is ready (clock stable after startup delay or PLL locked). If a clock source, which is not yet ready is selected, the switch occurs when the clock source becomes ready. Status bits in the [Clock control register \(RCC_CR\)](#) indicate which clock(s) is (are) ready and which clock is currently used as a system clock.

9.2.7 Clock security system (CSS)

Clock Security System can be activated by software. In this case, the clock detector is enabled after the HSE oscillator startup delay, and disabled when this oscillator is stopped.

If a failure is detected on the HSE clock, the HSE oscillator is automatically disabled, a clock failure event is sent to the break input of the advanced-control timers (TIM1/TIM8 and TIM15/16/17) and an interrupt is generated to inform the software about the failure (Clock Security System Interrupt CSSI), allowing the MCU to perform rescue operations. The CSSI is linked to the Cortex-M4[®] NMI (non-maskable interrupt) exception vector.

Note: Once the CSS is enabled and if the HSE clock fails, the CSS interrupt occurs and an NMI is automatically generated. The NMI is executed indefinitely unless the CSS interrupt pending bit is cleared. As a consequence, in the NMI ISR user must clear the CSS interrupt by setting the CSSC bit in the [Clock interrupt register \(RCC_CIR\)](#).

If the HSE oscillator is used directly or indirectly as the system clock (indirectly means: it is used as PLL input clock, and the PLL clock is used as system clock), a detected failure

causes a switch of the system clock to the HSI oscillator and the disabling of the HSE oscillator. If the HSE clock (divided or not) is the clock entry of the PLL used as system clock when the failure occurs, the PLL is disabled too.

9.2.8 ADC clock

The ADC clock is derived from the PLL output. It can reach 72 MHz and can be divided by the following prescalers values: 1, 2, 4, 6, 8, 10, 12, 16, 32, 64, 128 or 256. It is asynchronous to the AHB clock. Alternatively, the ADC clock can be derived from the AHB clock of the ADC bus interface, divided by a programmable factor (1, 2 or 4). This programmable factor is configured using the CKMODE bit fields in the ADCx_CCR.

If the programmed factor is '1', the AHB prescaler must be set to '1'.

9.2.9 RTC clock

The RTCCLK clock source can be either the HSE/32, LSE or LSI clock. It is selected by programming the RTCSEL[1:0] bits in the [RTC domain control register \(RCC_BDCR\)](#). This selection cannot be modified without resetting the RTC domain. The system must always be configured so as to get a PCLK frequency greater than or equal to the RTCCLK frequency for a proper operation of the RTC.

The LSE clock is in the RTC domain, whereas the HSE and LSI clocks are not. Consequently:

- If LSE is selected as RTC clock:
 - The RTC continues to work even if the V_{DD} supply is switched off, provided the V_{BAT} supply is maintained.
 - The RTC remains clocked and functional under system reset.
- If LSI is selected as the RTC clock:
 - The RTC state is not guaranteed if the V_{DD} supply is powered off.
- If the HSE clock divided by 32 is used as the RTC clock:
 - The RTC state is not guaranteed if the V_{DD} supply is powered off or if the internal voltage regulator is powered off (removing power from the 1.8 V domain).

9.2.10 Timers (TIMx) clock

APB clock source

The timers clock frequencies are automatically defined by hardware. There are two cases:

1. If the APB prescaler equals 1, the timer clock frequencies are set to the same frequency as that of the APB domain.
2. Otherwise, they are set to twice ($\times 2$) the frequency of the APB domain.

PLL clock source

A clock issued from the PLL (PLLCLKx2) can be selected for TIMx ($x = 1, 8$ on the STM32F303xB/C and STM32F358xC; $x = 1, 2, 3, 4, 8, 15, 16, 17, 20$ on STM32F303xD/E; $x = 1$ on the STM32F303x6/8 and STM32F328x8). This configuration allows to feed TIMx with a frequency up to 144 MHz when the system clock source is the PLL.

In this configuration:

- On the STM32F303xB/C and STM32F358xC, AHB and APB2 prescalers are set to 1, that is, AHB and APB2 clocks are not divided with respect to the system clock.
- On the STM32F303xD/E, STM32F303x6/8 and STM32F328x8 AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively with respect to the system clock.

9.2.11 Watchdog clock

If the Independent watchdog (IWDG) is started by either hardware option or software access, the LSI oscillator is forced ON and cannot be disabled. After the LSI oscillator temporization, the clock is provided to the IWDG.

9.2.12 I2S clock (only in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE)

The I2S clock can be either the System clock or an external clock provided on I2S_CKIN pin. The selection of the I2S clock source is performed using bit 23 (I2SSRC) of RCC_CFGR register.

9.2.13 Clock-out capability

The microcontroller clock output (MCO) capability allows the clock to be output onto the external MCO pin. The configuration registers of the corresponding GPIO port must be programmed in alternate function mode. One of 5 clock signals can be selected as the MCO clock.

- LSI
- LSE
- SYSCLK
- HSI
- HSE
- PLL clock divided by 2

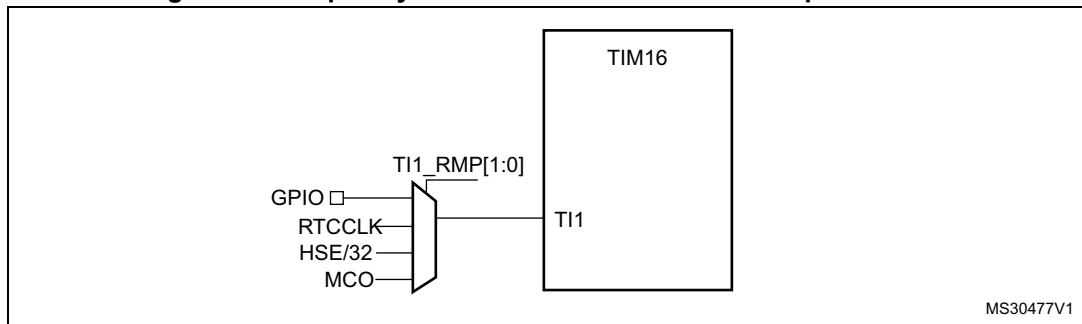
The selection is controlled by the MCO[2:0] bits in the [Clock configuration register \(RCC_CFGR\)](#).

On the STM32F303xD/E, STM32F303x6/8 and STM32F328x8, the additional bit PLLNODIV in this register controls the divider bypass for a PLL clock input to MCO. The MCO frequency can be reduced by a configurable divider, controlled by the MCOPRE[2:0] bits of the Clock configuration register (RCC_CFGR).

9.2.14 Internal/external clock measurement with TIM16

It is possible to indirectly measure the frequency of all on-board clock sources by means of the TIM16 channel 1 input capture. As represented on [Figure 17](#).

Figure 17. Frequency measurement with TIM16 in capture mode



The input capture channel of the Timer 16 can be a GPIO line or an internal clock of the MCU. This selection is performed through the T11_RMP [1:0] bits in the TIM16_OR register. The possibilities available are the following ones.

- TIM16 Channel1 is connected to the GPIO. Refer to the alternate function mapping in the device datasheets.
- TIM16 Channel1 is connected to the RTCCLK.
- TIM16 Channel1 is connected to the HSE/32 Clock.
- TIM16 Channel1 is connected to the microcontroller clock output (MCO), this selection is controlled by the MCO[2:0] bits of the Clock configuration register (RCC_CFGR).

Calibration of the HSI

The primary purpose of connecting the LSE, through the MCO multiplexer, to the channel 1 input capture is to be able to precisely measure the HSI system clocks (for this, the HSI should be used as the system clock source). The number of HSI clock counts between consecutive edges of the LSE signal provides a measure of the internal clock period. Taking advantage of the high precision of LSE crystals (typically a few tens of ppm's), it is possible to determine the internal clock frequency with the same resolution, and trim the source to compensate for manufacturing-process- and/or temperature- and voltage-related frequency deviations.

The HSI oscillator has dedicated user-accessible calibration bits for this purpose.

The basic concept consists in providing a relative measurement (for example, the HSI/LSE ratio): the precision is therefore closely related to the ratio between the two clock sources. The higher the ratio is, the better the measurement is.

If LSE is not available, HSE/32 is the better option to reach the most precise calibration possible.

Calibration of the LSI

The calibration of the LSI follows the same pattern that for the HSI, but changing the reference clock. It is necessary to connect LSI clock to the channel 1 input capture of the TIM16. Then define the HSE as system clock source, the number of his clock counts between consecutive edges of the LSI signal provides a measure of the internal low speed clock period.

The basic concept consists in providing a relative measurement (for example, the HSE/LSI ratio): the precision is therefore closely related to the ratio between the two clock sources. The higher the ratio is, the better the measurement is.

9.3 Low-power modes

APB peripheral clocks and DMA clock can be disabled by software.

Sleep mode stops the CPU clock. The memory interface clocks (Flash and RAM interfaces) can be stopped by software during sleep mode. The AHB to APB bridge clocks are disabled by hardware during Sleep mode when all the clocks of the peripherals connected to them are disabled.

Stop mode stops all the clocks in the V18 domain and disables the PLL, the HSI and the HSE oscillators.

All U(S)ARTs and I2Cs have the capability to enable the HSI oscillator even when the MCU is in Stop mode (if HSI is selected as the clock source for that peripheral).

All U(S)ARTs can also be driven by the LSE oscillator when the system is in Stop mode (if LSE is selected as clock source for that peripheral) and the LSE oscillator is enabled (LSEON) but they do not have the capability to turn on the LSE oscillator.

Standby mode stops all the clocks in the V18 domain and disables the PLL and the HSI and HSE oscillators.

The CPU's deepsleep mode can be overridden for debugging by setting the DBG_STOP or DBG_STANDBY bits in the DBGMCU_CR register.

When waking up from deepsleep after an interrupt (Stop mode) or reset (Standby mode), the HSI oscillator is selected as system clock.

If a flash programming operation is on going, deep-sleep mode entry is delayed until the flash interface access is finished. If an access to the APB domain is ongoing, deepsleep mode entry is delayed until the APB access is finished.

9.4 RCC registers

Refer to [Section 2.2](#) for a list of abbreviations used in register descriptions.

9.4.1 Clock control register (RCC_CR)

Address offset: 0x00

Reset value: 0x0000 XX83

(where X is undefined)

Access: no wait state, word, half-word, and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	PLL RDY	PLLON	Res.	Res.	Res.	Res.	CSS ON	HSE BYP	HSE RDY	HSE ON
						r	rw					rw	rw	r	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]					Res.	HSI RDY	HSION
r	r	r	r	r	r	r	r	rw	rw	rw	rw	rw		r	rw

Bits 31:26 Reserved, must be kept at reset value.

Bit 25 **PLL RDY**: PLL clock ready flag

Set by hardware to indicate that the PLL is locked.

0: PLL unlocked

1: PLL locked

Bit 24 **PLLON**: PLL enable

Set and cleared by software to enable PLL.

Cleared by hardware when entering Stop or Standby mode. This bit can not be reset if the PLL clock is used as system clock or is selected to become the system clock.

0: PLL OFF

1: PLL ON

Bits 23:20 Reserved, must be kept at reset value.

Bit 19 **CSSON**: Clock security system enable

Set and cleared by software to enable the clock security system. When CSSON is set, the clock detector is enabled by hardware when the HSE oscillator is ready, and disabled by hardware if a HSE clock failure is detected.

0: Clock detector OFF

1: Clock detector ON (Clock detector ON if the HSE oscillator is ready, OFF if not).

Bit 18 **HSEBYP**: HSE crystal oscillator bypass

Set and cleared by software to bypass the oscillator with an external clock. The external clock must be enabled with the HSEON bit set, to be used by the device. The HSEBYP bit can be written only if the HSE oscillator is disabled.

0: HSE crystal oscillator not bypassed

1: HSE crystal oscillator bypassed with external clock

- Bit 17 **HSERDY**: HSE clock ready flag
Set by hardware to indicate that the HSE oscillator is stable. This bit needs 6 cycles of the HSE oscillator clock to fall down after HSEON reset.
0: HSE oscillator not ready
1: HSE oscillator ready
- Bit 16 **HSEON**: HSE clock enable
Set and cleared by software.
Cleared by hardware to stop the HSE oscillator when entering Stop or Standby mode. This bit cannot be reset if the HSE oscillator is used directly or indirectly as the system clock.
0: HSE oscillator OFF
1: HSE oscillator ON
- Bits 15:8 **HSICAL[7:0]**: HSI clock calibration
These bits are initialized automatically at startup.
- Bits 7:3 **HSITRIM[4:0]**: HSI clock trimming
These bits provide an additional user-programmable trimming value that is added to the HSICAL[7:0] bits. It can be programmed to adjust to variations in voltage and temperature that influence the frequency of the HSI.
The default value is 16, which, when added to the HSICAL value, should trim the HSI to 8 MHz $\pm 1\%$. The trimming step (F_{hstrim}) is around 40 kHz between two consecutive HSICAL steps.
- Bit 2 Reserved, must be kept at reset value.
- Bit 1 **HSIRDY**: HSI clock ready flag
Set by hardware to indicate that HSI oscillator is stable. After the HSION bit is cleared, HSIRDY goes low after 6 HSI oscillator clock cycles.
0: HSI oscillator not ready
1: HSI oscillator ready
- Bit 0 **HSION**: HSI clock enable
Set and cleared by software.
Set by hardware to force the HSI oscillator ON when leaving Stop or Standby mode or in case of failure of the HSE crystal oscillator used directly or indirectly as system clock. This bit cannot be reset if the HSI is used directly or indirectly as system clock or is selected to become the system clock.
0: HSI oscillator OFF
1: HSI oscillator ON

9.4.2 Clock configuration register (RCC_CFGR)

Address offset: 0x04

Reset value: 0x0000 0000

Access: $0 \leq \text{wait state} \leq 2$, word, half-word and byte access

1 or 2 wait states inserted only if the access occurs during clock source switch.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PLLNO DIV	MCOPRE[2:1]		MCOF/ MCOP RE0	Res	MCO[2:0]			I2SSSRC	USBPR E	PLLMUL[3:0]				PLL XTPRE	PLL SRC
rw	rw	rw	r / rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PLLSR C ⁽¹⁾	Res.	PPRE2[2:0]			PPRE1[2:0]			HPRE[3:0]				SWS[1:0]		SW[1:0]	
rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	r	r	rw	rw

1. STM32F303xD/E only

Bit 31 **PLLNO DIV**: Do not divide PLL to MCO (in STM32F303x6/8 and STM32F328x8 , STM32F303xDxE and STM32F398xE only)
 This bit is set and cleared by software. It switch-off divider-by-2 for PLL connection to MCO
 0: PLL is divided by 2 before MCO
 1: PLL is not divided before MCO

Bits 30:28 **MCOPRE[2:0]**: Microcontroller Clock Output Prescaler (in STM32F303x6/8 and STM32F328x8 , STM32F303xDxE and STM32F398xE only)
 These bits are set and cleared by software. It is highly recommended to change this prescaler before MCO output is enabled
 000: MCO is divided by 1
 001: MCO is divided by 2
 010: MCO is divided by 4

 111: MCO is divided by 128

Bit 28 **MCOF**: Microcontroller Clock Output Flag (STM32F303xB/C and STM32F358xC only)
 Set and reset by hardware.
 It is reset by hardware when MCO field is written with a new value
 It is set by hardware when the switch to the new MCO source is effective.

Bit 27 Reserved, must be kept at reset value.

Bits 26:24 **MCO[2:0]**: Microcontroller clock output

Set and cleared by software.

000: MCO output disabled, no clock on MCO

001: Reserved

010: LSI clock selected.

011: LSE clock selected.

100: System clock (SYSCLK) selected

101: HSI clock selected

110: HSE clock selected

111: PLL clock selected (divided by 1 or 2 depending on PLLNODIV bit).

Note: This clock output may have some truncated cycles at startup or during MCO clock source switching.

Bit 23 **I2SSRC**: I2S external clock source selection (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

Set and reset by software to clock I2S2 and I2S3 with an external clock. This bits must be valid before enabling I2S2-3 clocks.

0: I2S2 and I2S3 clocked by system clock

1: I2S2 and I2S3 clocked by the external clock

Bit 22 **USBPRES**: USB prescaler

This bit is set and reset by software to generate the 48 MHz USB clock. They must be valid before enabling USB clocks.

0: PLL clock is divided by 1.5

1: PLL clock is not divided

Bits 21:18 **PLLMUL[3:0]**: PLL multiplication factor

These bits are written by software to define the PLL multiplication factor. These bits can be written only when PLL is disabled.

Caution: The PLL output frequency must not exceed 72 MHz.

0000: PLL input clock x 2

0001: PLL input clock x 3

0010: PLL input clock x 4

0011: PLL input clock x 5

0100: PLL input clock x 6

0101: PLL input clock x 7

0110: PLL input clock x 8

0111: PLL input clock x 9

1000: PLL input clock x 10

1001: PLL input clock x 11

1010: PLL input clock x 12

1011: PLL input clock x 13

1100: PLL input clock x 14

1101: PLL input clock x 15

1110: PLL input clock x 16

1111: PLL input clock x 16

- Bit 17 **PLLXTPRE**: HSE divider for PLL input clock
 This bit is set and cleared by software to select the HSE division factor for the PLL. It can be written only when the PLL is disabled.
Note: This bit is the same as the LSB of PREDIV in [Clock configuration register 2 \(RCC_CFGR2\)](#) (for compatibility with other STM32 products)
 0000: HSE input to PLL not divided
 0001: HSE input to PLL divided by 2
- Bits 16:15 **PLL SRC**: PLL entry clock source (STM32F303xD/E and STM32F398xE only)
 Set and cleared by software to select PLL clock source. These bits can be written only when PLL is disabled.
 00: HSI/2 used as PREDIV1 entry and PREDIV1 forced to div by 2.
 01: HSI used as PREDIV1 entry.
 10: HSE used as PREDIV1 entry.
 11: Reserved.
- Bit 16 **PLL SRC**: PLL entry clock source (in STM32F303xB/C and STM32F358xC and STM32F303x6/8 and STM32F328x8 devices)
 Set and cleared by software to select PLL clock source. This bit can be written only when PLL is disabled.
 0: HSI/2 selected as PLL input clock
 1: HSE/PREDIV selected as PLL input clock (refer to [Section 9.4.12: Clock configuration register 2 \(RCC_CFGR2\)](#) on page 162)
- Bit 15 Reserved, must be kept at reset value in STM32F303xB/C, STM32F358xC, STM32F303x6/8 and STM32F328x8 devices, and used with Bit 16 in STM32F303xD/E to select the PLL clock source.
- Bit 14 Reserved, must be kept at reset value.
- Bits 13:11 **PPRE2[2:0]**: APB high-speed prescaler (APB2)
 Set and cleared by software to control the division factor of the APB2 clock (PCLK).
 0xx: HCLK not divided
 100: HCLK divided by 2
 101: HCLK divided by 4
 110: HCLK divided by 8
 111: HCLK divided by 16
- Bits 10:8 **PPRE1[2:0]**: APB Low-speed prescaler (APB1)
 Set and cleared by software to control the division factor of the APB1 clock (PCLK).
 0xx: HCLK not divided
 100: HCLK divided by 2
 101: HCLK divided by 4
 110: HCLK divided by 8
 111: HCLK divided by 16

Bits 7:4 **HPRE[3:0]**: HCLK prescaler

Set and cleared by software to control the division factor of the AHB clock.

0xxx: SYSCLK not divided
 1000: SYSCLK divided by 2
 1001: SYSCLK divided by 4
 1010: SYSCLK divided by 8
 1011: SYSCLK divided by 16
 1100: SYSCLK divided by 64
 1101: SYSCLK divided by 128
 1110: SYSCLK divided by 256
 1111: SYSCLK divided by 512

Note: The prefetch buffer must be kept on when using a prescaler different from 1 on the AHB clock. Refer to section [Read operations on page 69](#) for more details.

Bits 3:2 **SWS[1:0]**: System clock switch status

Set and cleared by hardware to indicate which clock source is used as system clock.

00: HSI oscillator used as system clock
 01: HSE oscillator used as system clock
 10: PLL used as system clock
 11: not applicable

Bits 1:0 **SW[1:0]**: System clock switch

Set and cleared by software to select SYSCLK source.

Cleared by hardware to force HSI selection when leaving Stop and Standby mode or in case of failure of the HSE oscillator used directly or indirectly as system clock (if the Clock Security System is enabled).

00: HSI selected as system clock
 01: HSE selected as system clock
 10: PLL selected as system clock
 11: not allowed

9.4.3 Clock interrupt register (RCC_CIR)

Address offset: 0x08

Reset value: 0x0000 0000

Access: no wait state, word, half-word, and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CSSC	Res.	Res.	PLL RDYC	HSE RDYC	HSI RDYC	LSE RDYC	LSI RDYC
								w			w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	PLL RDYIE	HSE RDYIE	HSI RDYIE	LSE RDYIE	LSI RDYIE	CSSF	Res.	Res.	PLL RDYF	HSE RDYF	HSI RDYF	LSE RDYF	LSI RDYF
			rw	rw	rw	rw	rw	r			r	r	r	r	r

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **CSSC**: Clock security system interrupt clear

This bit is set by software to clear the CSSF flag.

0: No effect

1: Clear CSSF flag

Bits 22:21 Reserved, must be kept at reset value.

Bit 20 **PLLRDYC**: PLL ready interrupt clear

This bit is set by software to clear the PLLRDYF flag.

0: No effect

1: Clear PLLRDYF flag

Bit 19 **HSERDYC**: HSE ready interrupt clear

This bit is set by software to clear the HSERDYF flag.

0: No effect

1: Clear HSERDYF flag

Bit 18 **HSIRDYC**: HSI ready interrupt clear

This bit is set software to clear the HSIRDYF flag.

0: No effect

1: Clear HSIRDYF flag

Bit 17 **LSERDYC**: LSE ready interrupt clear

This bit is set by software to clear the LSERDYF flag.

0: No effect

1: LSERDYF cleared

Bit 16 **LSIRDYC**: LSI ready interrupt clear

This bit is set by software to clear the LSIRDYF flag.

0: No effect

1: LSIRDYF cleared

Bits 15:13 Reserved, must be kept at reset value.

Bit 12 **PLLRDYIE**: PLL ready interrupt enable

Set and cleared by software to enable/disable interrupt caused by PLL lock.

0: PLL lock interrupt disabled

1: PLL lock interrupt enabled

Bit 11 **HSERDYIE**: HSE ready interrupt enable

Set and cleared by software to enable/disable interrupt caused by the HSE oscillator stabilization.

0: HSE ready interrupt disabled

1: HSE ready interrupt enabled

Bit 10 **HSIRDYIE**: HSI ready interrupt enable

Set and cleared by software to enable/disable interrupt caused by the HSI oscillator stabilization.

0: HSI ready interrupt disabled

1: HSI ready interrupt enabled

- Bit 9 **LSERDYIE**: LSE ready interrupt enable
Set and cleared by software to enable/disable interrupt caused by the LSE oscillator stabilization.
0: LSE ready interrupt disabled
1: LSE ready interrupt enabled
- Bit 8 **LSIRDYIE**: LSI ready interrupt enable
Set and cleared by software to enable/disable interrupt caused by the LSI oscillator stabilization.
0: LSI ready interrupt disabled
1: LSI ready interrupt enabled
- Bit 7 **CSSF**: Clock security system interrupt flag
Set by hardware when a failure is detected in the HSE oscillator.
Cleared by software setting the CSSC bit.
0: No clock security interrupt caused by HSE clock failure
1: Clock security interrupt caused by HSE clock failure
- Bits 6:5 Reserved, must be kept at reset value.
- Bit 4 **PLLRDYF**: PLL ready interrupt flag
Set by hardware when the PLL locks and PLLRDYE is set.
Cleared by software setting the PLLRDYC bit.
0: No clock ready interrupt caused by PLL lock
1: Clock ready interrupt caused by PLL lock
- Bit 3 **HSERDYF**: HSE ready interrupt flag
Set by hardware when the HSE clock becomes stable and HSERDYE is set.
Cleared by software setting the HSERDYC bit.
0: No clock ready interrupt caused by the HSE oscillator
1: Clock ready interrupt caused by the HSE oscillator
- Bit 2 **HSIRDYF**: HSI ready interrupt flag
Set by hardware when the HSI clock becomes stable and HSIRDYE is set in a response to setting the HSION (refer to [Clock control register \(RCC_CR\)](#)). When HSION is not set but the HSI oscillator is enabled by the peripheral through a clock request, this bit is not set and no interrupt is generated.
Cleared by software setting the HSIRDYC bit.
0: No clock ready interrupt caused by the HSI oscillator
1: Clock ready interrupt caused by the HSI oscillator
- Bit 1 **LSERDYF**: LSE ready interrupt flag
Set by hardware when the LSE clock becomes stable and LSERDYE is set.
Cleared by software setting the LSERDYC bit.
0: No clock ready interrupt caused by the LSE oscillator
1: Clock ready interrupt caused by the LSE oscillator
- Bit 0 **LSIRDYF**: LSI ready interrupt flag
Set by hardware when the LSI clock becomes stable and LSIRDYE is set.
Cleared by software setting the LSIRDYC bit.
0: No clock ready interrupt caused by the LSI oscillator
1: Clock ready interrupt caused by the LSI oscillator

9.4.4 APB2 peripheral reset register (RCC_APB2RSTR)

Address offset: 0x0C

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIM20 RST	Res.	TIM17 RST	TIM16 RST	TIM15 RST
													rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI4RST	USART1 RST	TIM8 RST	SPI1 RST	TIM1 RST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYS CFG RST
rw	rw	rw	rw	rw											rw

Bits 31:2119 Reserved, must be kept at reset value.

Bit 20 **TIM20RST**: TIM20 timer reset (only on STM32F303xD/E and STM32F398xE devices)

Set and cleared by software.

0: No effect

1: Reset TIM20 timer

Bit 19 Reserved, must be kept at reset value.

Bit 18 **TIM17RST**: TIM17 timer reset

Set and cleared by software.

0: No effect

1: Reset TIM17 timer

Bit 17 **TIM16RST**: TIM16 timer reset

Set and cleared by software.

0: No effect

1: Reset TIM16 timer

Bit 16 **TIM15RST**: TIM15 timer reset

Set and cleared by software.

0: No effect

1: Reset TIM15 timer

Bit 15 **SPI4RST**: SPI4 reset (only on STM32F303xD/E and STM32F398xE devices)

Set and cleared by software.

0: No effect

1: Reset SPI4

Bit 14 **USART1RST**: USART1 reset

Set and cleared by software.

0: No effect

1: Reset USART1

Bit 13 **TIM8RST**: TIM8 timer reset (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

Set and cleared by software.

0: No effect

1: Reset TIM8 timer

Bit 12 **SPI1RST**: SPI1 reset

Set and cleared by software.

0: No effect

1: Reset SPI1

Bit 11 **TIM1RST**: TIM1 timer reset

Set and cleared by software.

0: No effect

1: Reset TIM1 timer

Bits 10:1 Reserved, must be kept at reset value.

Bit 0 **SYSCFGRST**: SYSCFG, Comparators and operational amplifiers reset

Set and cleared by software.

0: No effect

1: Reset SYSCFG, COMP, and OPAMP

9.4.5 APB1 peripheral reset register (RCC_APB1RSTR)

Address offset: 0x10

Reset value: 0x0000 0000

Access: no wait state, word, half-word, and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	I2C3 RST ⁽¹⁾	DAC1 RST	PWR RST	Res.	DAC2R ST	CAN RST	Res.	USB RST	I2C2 RST	I2C1 RST	UART5 RST	UART4 RST	USART3 RST	USART2 RST	Res.
		rw	rw		rw	rw		rw	rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 RST	SPI2 RST	Res.	Res.	WWDG RST	Res.	Res.	Res.	Res.	Res.	TIM7 RST	TIM6 RST	Res.	TIM4 RST	TIM3 RST	TIM2 RST
rw	rw			rw						rw	rw		rw	rw	rw

1. STM32F303xD/E devices only.

Bit 31 Reserved, must be kept at reset value.

Bit 30 **I2C3RST**: I2C3 reset (STM32F303xD/E and STM32F398xE devices only)

Set and cleared by software.

0: No effect

1: Reset I2C3

Bit 29 **DAC1RST**: DAC1 interface reset

Set and cleared by software.

0: No effect

1: Reset DAC1 interface

Bit 28 **PWRRST**: Power interface reset

Set and cleared by software.

0: No effect

1: Reset power interface

Bit 27 Reserved, must be kept at reset value.

- Bit 26 **DAC2RST**: DAC2 interface reset (STM32F303x6/8 and STM32F328x8 devices only)
Set and cleared by software.
0: No effect
1: Reset DAC2 interface
- Bit 25 **CANRST**: CAN reset
Set and reset by software.
0: does not reset the CAN
1: resets the CAN
- Bit 24 Reserved, must be kept at reset value
- Bit 23 **USBRST**: USB reset (STM32F303xB/C/D/E devices only)
Set and reset by software.
0: does not reset USB
1: resets USB
- Bit 22 **I2C2RST**: I2C2 reset (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: No effect
1: Reset I2C2
- Bit 21 **I2C1RST**: I2C1 reset
Set and cleared by software.
0: No effect
1: Reset I2C1
- Bit 20 **UART5RST**: UART5 reset (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: No effect
1: Reset UART5
- Bit 19 **UART4RST**: UART4 reset (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: No effect
1: Reset UART4
- Bit 18 **USART3RST**: USART3 reset
Set and cleared by software.
0: No effect
1: Reset USART3
- Bit 17 **USART2RST**: USART2 reset
Set and cleared by software.
0: No effect
1: Reset USART2
- Bit 16 Reserved, must be kept at reset value.
- Bit 15 **SPI3RST**: SPI3 reset (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: No effect
1: Reset SPI3 and I2S3

Bit 14 **SPI2RST**: SPI2 reset (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.

0: No effect

1: Reset SPI2 and I2S2

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGRST**: Window watchdog reset
Set and cleared by software.

0: No effect

1: Reset window watchdog

Bits 10:6 Reserved, must be kept at reset value.

Bit 5 **TIM7RST**: TIM7 timer reset
Set and cleared by software.

0: No effect

1: Reset TIM7

Bit 4 **TIM6RST**: TIM6 timer reset
Set and cleared by software.

0: No effect

1: Reset TIM6

Bit 3 Reserved, must be kept at reset value.

Bit 2 **TIM4RST**: TIM4 timer reset (STM32F303xB/C and STM32F358xC devices only)
Set and cleared by software.

0: No effect

1: Reset TIM4

Bit 1 **TIM3RST**: TIM3 timer reset
Set and cleared by software.

0: No effect

1: Reset TIM3

Bit 0 **TIM2RST**: TIM2 timer reset
Set and cleared by software.

0: No effect

1: Reset TIM2

9.4.6 AHB peripheral clock enable register (RCC_AHBENR)

Address offset: 0x14

Reset value: 0x0000 0014

Access: no wait state, word, half-word and byte access

Note: *When the peripheral clock is not active, the peripheral register values may not be readable by software and the returned value is always 0x0.*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	ADC34 EN	ADC12EN	Res.	Res.	Res.	TSCEN	GPIO G EN ⁽¹⁾	GPIOF EN	GPIOE EN	PIOD EN	GPIOC EN	GPIOB EN	GPIOA EN	GPIOH EN ⁽¹⁾
		rw	rw				rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CRC EN	FMC EN ⁽¹⁾	FLITF EN	Res.	SRAM EN	DMA2 EN	DMA1 EN
									rw	rw	rw		rw	rw	rw

1. Only on STM32F303xDxE.

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **ADC34EN**: ADC3 and ADC4 enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

Set and reset by software.

0: ADC3 and ADC4 clock disabled

1: ADC3 and ADC4 clock enabled

Bit 28 **ADC12EN**: ADC1 and ADC2 enable

Set and reset by software.

0: ADC1 and ADC2 clock disabled

1: ADC1 and ADC2 clock enabled

Bits 27:25 Reserved, must be kept at reset value.

Bit 24 **TSCEN**: Touch sensing controller clock enable

Set and cleared by software.

0: TSC clock disabled

1: TSC clock enabled

Bit 23 **GPIOGEN**: IO port G clock enable. (Only on STM32F303xDxE)

Set and cleared by software.

0: IO port G clock disabled

1: IO port G clock enabled

Bit 22 **GPIOFEN**: I/O port F clock enable

Set and cleared by software.

0: I/O port F clock disabled

1: I/O port F clock enabled

Bit 21 **GPIOEEN**: I/O port E clock enable(STM32F303xB/C and STM32F358xC devices only)

Set and cleared by software.

0: I/O port E clock disabled

1: I/O port E clock enabled.

- Bit 20 **GPIODEN**: I/O port D clock enable
Set and cleared by software.
0: I/O port D clock disabled
1: I/O port D clock enabled
- Bit 19 **GPIOCEN**: I/O port C clock enable
Set and cleared by software.
0: I/O port C clock disabled
1: I/O port C clock enabled
- Bit 18 **GPIOBEN**: I/O port B clock enable
Set and cleared by software.
0: I/O port B clock disabled
1: I/O port B clock enabled
- Bit 17 **GPIOAEN**: I/O port A clock enable
Set and cleared by software.
0: I/O port A clock disabled
1: I/O port A clock enabled
- Bit 16 **GPIOHEN**: IO port H clock enable. (Only on STM32F303xDxE)
Set and cleared by software.
0: IO port H clock disabled
1: IO port H clock enabled
- Bits 15:7 Reserved, must be kept at reset value.
- Bit 6 **CRCEN**: CRC clock enable
Set and cleared by software.
0: CRC clock disabled
1: CRC clock enabled
- Bit 5 **FMCEN**: FMC clock enable. (Only on STM32F303xDxE)
Set and cleared by software.
0: FMC clock disabled
1: FMC clock enabled
- Bit 4 **FLITFEN**: FLITF clock enable
Set and cleared by software to disable/enable FLITF clock during Sleep mode.
0: FLITF clock disabled during Sleep mode
1: FLITF clock enabled during Sleep mode
- Bit 3 Reserved, must be kept at reset value.

Bit 2 **SRAMEN**: SRAM interface clock enable

Set and cleared by software to disable/enable SRAM interface clock during Sleep mode.

0: SRAM interface clock disabled during Sleep mode.

1: SRAM interface clock enabled during Sleep mode

Bit 1 **DMA2EN**: DMA2 clock enable (STM32F303xB/C and STM32F358xC devices only)

Set and cleared by software.

0: DMA2 clock disabled

1: DMA2 clock enabled

Bit 0 **DMA1EN**: DMA1 clock enable

Set and cleared by software.

0: DMA1 clock disabled

1: DMA1 clock enabled

9.4.7 APB2 peripheral clock enable register (RCC_APB2ENR)

Address: 0x18

Reset value: 0x0000 0000

Access: word, half-word, and byte access

No wait states, except if the access occurs while an access to a peripheral in the APB2 domain is on going. In this case, wait states are inserted until the access to APB2 peripheral is finished.

Note: *When the peripheral clock is not active, the peripheral register values may not be readable by software and the returned value is always 0x0.*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIM20 EN	Res.	TIM17 EN	TIM16 EN	TIM15 EN
											rw		rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI4EN	USART1EN	TIM8 EN	SPI1 EN	TIM1 EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYS CFGEN
rw	rw	rw	rw	rw											rw

Bits 31:21 Reserved, must be kept at reset value.

Bit 20 **TIM20EN**: TIM20 timer clock enable (STM32F303xD/E and STM32F398xE only)

Set and cleared by software.

0: TIM20 timer clock disabled

1: TIM20 timer clock enabled

Bit 19 Reserved, must be kept at reset value.

Bit 18 **TIM17EN**: TIM17 timer clock enable

Set and cleared by software.

0: TIM17 timer clock disabled

1: TIM17 timer clock enabled

- Bit 17 **TIM16EN**: TIM16 timer clock enable
Set and cleared by software.
0: TIM16 timer clock disabled
1: TIM16 timer clock enabled
- Bit 16 **TIM15EN**: TIM15 timer clock enable
Set and cleared by software.
0: TIM15 timer clock disabled
1: TIM15 timer clock enabled
- Bit 15 **SPI4EN**: SPI4 clock enable (STM32F303xD/E and STM32F398xE only)
Set and cleared by software.
0: SPI4 clock disabled
1: SPI4 clock enabled
- Bit 14 **USART1EN**: USART1 clock enable
Set and cleared by software.
0: USART1 clock disabled
1: USART1 clock enabled
- Bit 13 **TIM8EN**: TIM8 timer clock enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: TIM8 timer clock disabled
1: TIM8 timer clock enabled
- Bit 12 **SPI1EN**: SPI1 clock enable
Set and cleared by software.
0: SPI1 clock disabled
1: SPI1 clock enabled
- Bit 11 **TIM1EN**: TIM1 timer clock enable
Set and cleared by software.
0: TIM1 timer clock disabled
1: TIM1 timer clock enabled
- Bits 10:1 Reserved, must be kept at reset value.
- Bit 0 **SYSCFGEN**: COMP and SYSCFG clock enable.
Set and cleared by software.
0: SYSCFG clock disabled
1: SYSCFG clock enabled

9.4.8 APB1 peripheral clock enable register (RCC_APB1ENR)

Address: 0x1C

Reset value: 0x0000 0000

Access: word, half-word, and byte access

No wait state, except if the access occurs while an access to a peripheral on APB1 domain is on going. In this case, wait states are inserted until this access to APB1 peripheral is finished.

Note: *When the peripheral clock is not active, the peripheral register values may not be readable by software and the returned value is always 0x0.*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	I2C3 EN	DAC1 EN	PWR EN	Res.	DAC2 EN	CAN EN	Res.	USB EN	I2C2 EN	I2C1 EN	UART5 EN	UART4 EN	USART3 EN	USART2 EN	Res.
		rw	rw		rw	rw		rw	rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 EN	SPI2 EN	Res.	Res.	WWD GEN	Res.	Res.	Res.	Res.	Res.	TIM7E N	TIM6EN	Res.	TIM4EN	TIM3EN	TIM2 EN
rw	rw			rw						rw	rw		rw	rw	rw

Bit 31 Reserved, must be kept at reset value.

Bit 30 **I2C3EN**: I2C3 clock enable (only in STM32F303xD/E devices)

Set and cleared by software.

0: I2C3 clock disabled

1: I2C3 clock enabled

Bit 29 **DAC1EN**: DAC1 interface clock enable

Set and cleared by software.

0: DAC1 interface clock disabled

1: DAC1 interface clock enabled

Bit 28 **PWREN**: Power interface clock enable

Set and cleared by software.

0: Power interface clock disabled

1: Power interface clock enabled

Bit 27 Reserved, must be kept at reset value.

Bit 26 **DAC2EN**: DAC2 interface clock enable (STM32F303x6/8 and STM32F328x8 devices only)

Set and cleared by software.

0: DAC2 interface clock disabled

1: DAC2 interface clock enabled

Bit 25 **CANEN**: CAN clock enable

Set and reset by software.

0: CAN clock disabled

1: CAN clock enabled

Bit 24 Reserved, must be kept at reset value.

- Bit 23 **USBEN**: USB clock enable (STM32F303xB/C/D/E devices only)
Set and reset by software.
0: USB clock disabled
1: USB clock enabled
- Bit 22 **I2C2EN**: I2C2 clock enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: I2C2 clock disabled
1: I2C2 clock enabled
- Bit 21 **I2C1EN**: I2C1 clock enable
Set and cleared by software.
0: I2C1 clock disabled
1: I2C1 clock enabled
- Bit 20 **UART5EN**: UART5 clock enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: UART5 clock disabled
1: UART5 clock enabled
- Bit 19 **UART4EN**: UART4 clock enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: UART4 clock disabled
1: UART4 clock enabled
- Bit 18 **USART3EN**: USART3 clock enable
Set and cleared by software.
0: USART3 clock disabled
1: USART3 clock enabled
- Bit 17 **USART2EN**: USART2 clock enable
Set and cleared by software.
0: USART2 clock disabled
1: USART2 clock enabled
- Bit 16 Reserved, must be kept at reset value.
- Bit 15 **SPI3EN**: SPI3 clock enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: SPI3 clock disabled
1: SPI3 clock enabled
- Bit 14 **SPI2EN**: SPI2 clock enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
Set and cleared by software.
0: SPI2 clock disabled
1: SPI2 clock enabled
- Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGEN**: Window watchdog clock enable

Set and cleared by software.

0: Window watchdog clock disabled

1: Window watchdog clock enabled

Bits 10:6 Reserved, must be kept at reset value.

Bit 5 **TIM7EN**: TIM7 timer clock enable

Set and cleared by software.

0: TIM7 clock disabled

1: TIM7 clock enabled

Bit 4 **TIM6EN**: TIM6 timer clock enable

Set and cleared by software.

0: TIM6 clock disabled

1: TIM6 clock enabled

Bit 3 Reserved, must be kept at reset value.

Bit 2 **TIM4EN**: TIM4 timer clock enable (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

Set and cleared by software.

0: TIM4 clock disabled

1: TIM4 clock enabled

Bit 1 **TIM3EN**: TIM3 timer clock enable

Set and cleared by software.

0: TIM3 clock disabled

1: TIM3 clock enabled

Bit 0 **TIM2EN**: TIM2 timer clock enable

Set and cleared by software.

0: TIM2 clock disabled

1: TIM2 clock enabled

9.4.9 RTC domain control register (RCC_BDCR)

Address offset: 0x20

Reset value: 0x0000 0018 (reset by RTC domain Reset)

Access: $0 \leq \text{wait state} \leq 3$, word, half-word, and byte access

Wait states are inserted in case of successive accesses to this register.

Note: The LSEON, LSEBYP, RTCSEL, and RTCEN bits of the [RTC domain control register \(RCC_BDCR\)](#) are in the RTC domain. As a result, after Reset, these bits are write-protected and the DBP bit in the [Power control register \(PWR_CR\)](#) has to be set before these can be modified. These bits are only reset after an RTC domain Reset (see [Section 9.1.3: RTC domain reset](#)). Any internal or external reset does not have any effect on these bits.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BDRST
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTC EN	Res.	Res.	Res.	Res.	Res.	RTCSEL[1:0]		Res.	Res.	Res.	LSEDRV[1:0]		LSE BYP	LSE RDY	LSEON
rw						rw	rw				rw	rw	rw	r	rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **BDRST**: RTC domain software reset

Set and cleared by software.

0: Reset not activated

1: Resets the entire RTC domain

Bit 15 **RTCEN**: RTC clock enable

Set and cleared by software.

0: RTC clock disabled

1: RTC clock enabled

Bits 14:10 Reserved, must be kept at reset value.

Bits 9:8 **RTCSEL[1:0]**: RTC clock source selection

Set by software to select the clock source for the RTC. Once the RTC clock source has been selected, it cannot be changed anymore unless the RTC domain is reset. The BDRST bit can be used to reset them.

00: No clock

01: LSE oscillator clock used as RTC clock

10: LSI oscillator clock used as RTC clock

11: HSE oscillator clock divided by 32 used as RTC clock

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:3 **LSEDRV[1:0]**: LSE oscillator drive capability

Set and reset by software to modulate the LSE oscillator's drive capability. A reset of the RTC domain restores the default value.

00: 'Xtal mode' lower driving capability

01: 'Xtal mode' medium high driving capability

10: 'Xtal mode' medium low driving capability

11: 'Xtal mode' higher driving capability (reset value)

Note: The oscillator is in Xtal mode when it is not in bypass mode.

Bit 2 **LSEBYP**: LSE oscillator bypass

Set and cleared by software to bypass oscillator in debug mode. This bit can be written only when the external 32 kHz oscillator is disabled.

- 0: LSE oscillator not bypassed
- 1: LSE oscillator bypassed

Bit 1 **LSERDY**: LSE oscillator ready

Set and cleared by hardware to indicate when the external 32 kHz oscillator is stable. After the LSEON bit is cleared, LSERDY goes low after 6 external low-speed oscillator clock cycles.

- 0: LSE oscillator not ready
- 1: LSE oscillator ready

Bit 0 **LSEON**: LSE oscillator enable

Set and cleared by software.

- 0: LSE oscillator OFF
- 1: LSE oscillator ON

9.4.10 Control/status register (RCC_CSR)

Address: 0x24

Reset value: 0x0C00 0000 (reset by system Reset, except reset flags by power Reset only)

Access: 0 ≤ wait state ≤ 3, word, half-word, and byte access

Wait states are inserted in case of successive accesses to this register.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LPWR RSTF	WWDG STF	IW WDG RSTF	SFT RSTF	POR RSTF	PIN RSTF	OB LRSTF	RMVF	V18PW RRSTF	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r	r	r	r	r	r	r	r	r							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LSI RDY	LSION
														r	rw

Bit 31 **LPWRSTF**: Low-power reset flag

Set by hardware when a Low-power management reset occurs.
Cleared by writing to the RMVF bit.

- 0: No Low-power management reset occurred
- 1: Low-power management reset occurred

For further information on low-power management reset, refer to [Reset](#).

Bit 30 **WWDGRSTF**: Window watchdog reset flag

Set by hardware when a window watchdog reset occurs.
Cleared by writing to the RMVF bit.

- 0: No window watchdog reset occurred
- 1: Window watchdog reset occurred

Bit 29 **IWDGRSTF**: Independent window watchdog reset flag

Set by hardware when an independent watchdog reset from V_{DD} domain occurs. Cleared by writing to the RMVF bit.

- 0: No watchdog reset occurred
- 1: Watchdog reset occurred

- Bit 28 **SFTRSTF**: Software reset flag
Set by hardware when a software reset occurs. Cleared by writing to the RMVF bit.
0: No software reset occurred
1: Software reset occurred
- Bit 27 **PORRSTF**: POR/PDR flag
Set by hardware when a POR/PDR occurs. Cleared by writing to the RMVF bit.
0: No POR/PDR occurred
1: POR/PDR occurred
- Bit 26 **PINRSTF**: PIN reset flag
Set by hardware when a reset from the NRST pin occurs. Cleared by writing to the RMVF bit.
0: No reset from NRST pin occurred
1: Reset from NRST pin occurred
- Bit 25 **OBLRSTF**: Option byte loader reset flag
Set by hardware when a reset from the OBL occurs. Cleared by writing to the RMVF bit.
0: No reset from OBL occurred
1: Reset from OBL occurred
- Bit 24 **RMVF**: Remove reset flag
Set by software to clear the reset flags.
0: No effect
1: Clear the reset flags
- Bit 23 **V18PWRSTF**: Reset flag of the 1.8 V domain.
Set by hardware when a POR/PDR of the 1.8 V domain occurred. Cleared by writing to the RMVF bit.
0: No POR/PDR reset of the 1.8 V domain occurred
1: POR/PDR reset of the 1.8 V domain occurred
Note: On the STM32F3x8 products, this flag is reserved.
- Bits 22:2 Reserved, must be kept at reset value.
- Bit 1 **LSIRDY**: LSI oscillator ready
Set and cleared by hardware to indicate when the LSI oscillator is stable. After the LSION bit is cleared, LSIRDY goes low after 3 LSI oscillator clock cycles.
0: LSI oscillator not ready
1: LSI oscillator ready
- Bit 0 **LSION**: LSI oscillator enable
Set and cleared by software.
0: LSI oscillator OFF
1: LSI oscillator ON

9.4.11 AHB peripheral reset register (RCC_AHBRSTR)

Address: 0x28

Reset value: 0x0000 0000

Access: no wait states, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	ADC34 RST	ADC12 RST	Res.	Res.	Res.	TSC RST	GPIOG RST ⁽¹⁾	GPIOF RST	GPIOE RST	GPIOD RST	GPIOC RST	GPIOB RST	GPIOA RST	GPIOH RST ⁽¹⁾
		rw	rw				rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FMCR ST ⁽¹⁾	Res.	Res.	Res.	Res.	Res.
										rw					

1. Only on STM32F303xDxE.

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **ADC34RST**: ADC3 and ADC4 reset (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

Set and reset by software.

- 0: does not reset the ADC3 and ADC4
- 1: resets the ADC3 and ADC4

Bit 28 **ADC12RST**: ADC1 and ADC2 reset

Set and reset by software.

- 0: does not reset the ADC1 and ADC2
- 1: resets the ADC1 and ADC2

Bits 27:25 Reserved, must be kept at reset value.

Bit 24 **TSCRST**: Touch sensing controller reset

Set and cleared by software.

- 0: No effect
- 1: Reset TSC

Bit 23 **GPIOGRST**: I/O port G reset. (Only on STM32F303xDxE)

Set and cleared by software.

- 0: No effect
- 1: Reset I/O port G

Bit 22 **GPIOFRST**: I/O port F reset

Set and cleared by software.

- 0: No effect
- 1: Reset I/O port F

Bit 21 **GPIOERST**: I/O port E reset

Set and cleared by software.

- 0: No effect
- 1: Reset I/O port E

- Bit 20 **GPIODRST**: I/O port D reset
Set and cleared by software.
0: No effect
1: Reset I/O port D
- Bit 19 **GPIOCRST**: I/O port C reset
Set and cleared by software.
0: No effect
1: Reset I/O port C
- Bit 18 **GPIOBRST**: I/O port B reset
Set and cleared by software.
0: No effect
1: Reset I/O port B
- Bit 17 **GPIOARST**: I/O port A reset
Set and cleared by software.
0: No effect
1: Reset I/O port A
- Bit 16 **GPIOHRST**: I/O port H reset (Only on STM32F303xDxE).
Set and cleared by software.
0: No effect
1: Reset I/O port H

Bits 15:6 Reserved, must be kept at reset value.

- Bit 5 **FMCRST**: FMC reset (Only on STM32F303xDxE).
Set and cleared by software.
0: No effect
1: Reset FMC

Bits 4:0 Reserved, must be kept at reset value.

9.4.12 Clock configuration register 2 (RCC_CFGR2)

Address: 0x2C

Reset value: 0x0000 0000

Access: no wait states, word, half-word, and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	Res	ADC34PRES[4:0]					ADC12PRES[4:0]					PREDIV[3:0]			
		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 13:9 **ADC34PRES**: ADC34 prescaler (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE only)

Set and reset by software to control PLL clock to ADC34 division factor.

0xxxx: ADC34 clock disabled, ADC34 can use AHB clock

10000: PLL clock divided by 1

10001: PLL clock divided by 2

10010: PLL clock divided by 4

10011: PLL clock divided by 6

10100: PLL clock divided by 8

10101: PLL clock divided by 10

10110: PLL clock divided by 12

10111: PLL clock divided by 16

11000: PLL clock divided by 32

11001: PLL clock divided by 64

11010: PLL clock divided by 128

11011: PLL clock divided by 256

others: PLL clock divided by 256

Bits 8:4 **ADC12PRES**: ADC12 prescaler

Set and reset by software to control PLL clock to ADC12 division factor.

0xxxx: ADC12 clock disabled, ADC12 can use AHB clock

10000: PLL clock divided by 1

10001: PLL clock divided by 2

10010: PLL clock divided by 4

10011: PLL clock divided by 6

10100: PLL clock divided by 8

10101: PLL clock divided by 10

10110: PLL clock divided by 12

10111: PLL clock divided by 16

11000: PLL clock divided by 32

11001: PLL clock divided by 64

11010: PLL clock divided by 128

11011: PLL clock divided by 256

others: PLL clock divided by 256

Bits 3:0 **PREDIV**: PREDIV division factor

These bits are set and cleared by software to select PREDIV division factor. They can be written only when the PLL is disabled.

Note: Bit 0 is the same bit as bit 17 in [Clock configuration register \(RCC_CFGR\)](#), so modifying bit 17 [Clock configuration register \(RCC_CFGR\)](#) also modifies bit 0 in [Clock configuration register 2 \(RCC_CFGR2\)](#) (for compatibility with other STM32 products)

0000: HSE input to PLL not divided

0001: HSE input to PLL divided by 2

0010: HSE input to PLL divided by 3

0011: HSE input to PLL divided by 4

0100: HSE input to PLL divided by 5

0101: HSE input to PLL divided by 6

0110: HSE input to PLL divided by 7

0111: HSE input to PLL divided by 8

1000: HSE input to PLL divided by 9

1001: HSE input to PLL divided by 10

1010: HSE input to PLL divided by 11

1011: HSE input to PLL divided by 12

1100: HSE input to PLL divided by 13

1101: HSE input to PLL divided by 14

1110: HSE input to PLL divided by 15

1111: HSE input to PLL divided by 16

9.4.13 Clock configuration register 3 (RCC_CFGR3)

Address: 0x30

Reset value: 0x0000 0000

Access: no wait states, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	TIM34 SW ⁽¹⁾	TIM2 SW ⁽¹⁾	UART5SW[1:0]		UART4SW[1:0]		USART3SW[1:0] ⁽²⁾		USART2SW[1:0] ⁽²⁾	
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM20 SW ⁽¹⁾	Res.	TIM17 SW ⁽¹⁾	Res.	TIM16 SW ⁽¹⁾	TIM15 SW ⁽¹⁾	TIM8S W	TIM1 SW	Res.	I2C3 SW ⁽¹⁾	I2C2 SW	I2C1 SW	Res.	Res.	USART1SW[1:0]	
rw		rw		rw	rw	rw	rw		rw	rw	rw			rw	rw

1. Only on STM32F303xDxE.

2. Not available in STM32F303x6/8 and STM32F328x8 devices.

Bits 31:26 Reserved, must be kept at reset value.

Bit 25 **TIM34SW**: Timer34 clock source selection

Set and reset by software to select TIM34 clock source.

The bit is writable only when the following conditions occur: system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.

The bit is reset by hardware when exiting from the previous condition (user must set the bit again in case of a new switch is required)

0: PCLK2 clock (doubled frequency when prescaled) (default)

1: PLL vco output (running up to 144 MHz)

Note: STM32F303xDxE and STM32F398xE only.

Bit 24 **TIM2SW**: Timer2 clock source selection

Set and reset by software to select TIM2 clock source.

The bit is writable only when the following conditions occur: clock system = PLL, and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.

0: PCLK2 clock (doubled frequency when prescaled) (default)

1: PLL vco output (running up to 144 MHz)

Note: STM32F303xDxE and STM32F398xE only.

Bits 23:22 **UART5SW[1:0]**: UART5 clock source selection (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

This bit is set and cleared by software to select the UART5 clock source.

00: PCLK selected as UART5 clock source (default)

01: System clock (SYSCLK) selected as UART5 clock

10: LSE clock selected as UART5 clock

11: HSI clock selected as UART5 clock

Bits 21:20 **UART4SW[1:0]**: UART4 clock source selection (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)

This bit is set and cleared by software to select the UART4 clock source.

00: PCLK selected as UART4 clock source (default)

01: System clock (SYSCLK) selected as UART4 clock

10: LSE clock selected as UART4 clock

11: HSI clock selected as UART4 clock

Bits 19:18 **USART3SW[1:0]**: USART3 clock source selection

This bit is set and cleared by software to select the USART3 clock source.

00: PCLK selected as USART3 clock source (default)

01: System clock (SYSCLK) selected as USART3 clock

10: LSE clock selected as USART3 clock

11: HSI clock selected as USART3 clock

Note: USART2SW[1:0] is not available in STM32F303x6/8 and STM32F328x8.

Bits 17:16 **USART2SW[1:0]**: USART2 clock source selection

This bit is set and cleared by software to select the USART2 clock source.

00: PCLK selected as USART2 clock source (default)

01: System clock (SYSCLK) selected as USART2 clock

10: LSE clock selected as USART2 clock

11: HSI clock selected as USART2 clock

Note: USART2SW[1:0] is not available in STM32F303x6/8 and STM32F328x8.

Bit 15 TIM20SW: Timer20 clock source selection

Set and reset by software to select TIM20 clock source.

The bit is writable only when the following conditions occur: system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.

The bit is reset by hardware when exiting from the previous condition (user must set the bit again in case of a new switch is required)

0: PCLK2 clock (doubled frequency when prescaled) (default)

1: PLL vco output (running up to 144 MHz)

Note: STM32F303xDxE only.

Bit 14 Reserved, must be kept at reset value.**Bit 13 TIM17SW:** Timer17 clock source selection

Set and reset by software to select TIM17 clock source.

The bit is writable only when the following conditions occur: system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.

The bit is reset by hardware when exiting from the previous condition (user must set the bit again in case of a new switch is required)

0: PCLK2 clock (doubled frequency when prescaled) (default)

1: PLL vco output (running up to 144 MHz)

Note: STM32F303xDxE and STM32F398xE devices only.

Bit 12 Reserved, must be kept at reset value.**Bit 11 TIM16SW:** Timer16 clock source selection

Set and reset by software to select TIM16 clock source.

The bit is writable only when the following conditions occur: system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.

The bit is reset by hardware when exiting from the previous condition (user must set the bit again in case of a new switch is required)

0: PCLK2 clock (doubled frequency when prescaled) (default)

1: PLL vco output (running up to 144 MHz)

Note: STM32F303xD/E and STM32F398xE devices only.

Bit 10 TIM15SW: Timer15 clock source selection

Set and reset by software to select TIM15 clock source.

The bit is writable only when the following conditions occur: system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.

The bit is reset by hardware when exiting from the previous condition (user must set the bit again in case of a new switch is required)

0: PCLK2 clock (doubled frequency when prescaled) (default)

1: PLL vco output (running up to 144 MHz)

Note: STM32F303xD/E and STM32F398xE devices only.

- Bit 9 **TIM8SW**: Timer8 clock source selection (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
 Set and reset by software to select TIM8 clock source.
 In STM32F303xB/C and STM32F358xC, the bit is writable only when the following conditions occur: clock system = PLL, and AHB and APB2 subsystem clock not divided respect the clock system.
 In STM32F303xD/E and STM32F398xE, the bit is writable only when the following conditions occur: system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively..
 The bit is reset by hardware when exiting from the previous condition (user must set the bit again in case of a new switch is required)
 0: PCLK2 clock (doubled frequency when prescaled) (default)
 1: PLL vco output (running up to 144 MHz)
- Bit 8 **TIM1SW**: Timer1 clock source selection
 Set and reset by software to select TIM1 clock source.
 In STM32F303xB/C and STM32F358xC, the bit is writable only when the following conditions occur: clock system = PLL, and AHB and APB2 subsystem clock not divided respect the clock system.
 In STM32F303x6/8/D/E and STM32F398xE, the bit is writable only when the following conditions occur: system clock source is the PLL and AHB or APB2 subsystem clocks are not divided by more than 2 cumulatively.
 The bit is reset by hardware when exiting from the previous condition (user must set the bit again in case of a new switch is required)
 0: PCLK2 clock (doubled frequency when prescaled) (default)
 1: PLL vco output (running up to 144 MHz)
- Bit 7: Reserved, must be kept at reset value.
- Bit 6 **I2C3SW**: I2C3 clock source selection (STM32F303xD/E devices only)
 This bit is set and cleared by software to select the I2C3 clock source.
 0: HSI clock selected as I2C3 clock source (default)
 1: SYSCLK clock selected as I2C3 clock
- Bit 5 **I2C2SW**: I2C2 clock source selection (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices only)
 This bit is set and cleared by software to select the I2C2 clock source.
 0: HSI clock selected as I2C2 clock source (default)
 1: SYSCLK clock selected as I2C2 clock
- Bit 4 **I2C1SW**: I2C1 clock source selection
 This bit is set and cleared by software to select the I2C1 clock source.
 0: HSI clock selected as I2C1 clock source (default)
 1: SYSCLK clock selected as I2C1 clock
- Bits 3:2 Reserved, must be kept at reset value.
- Bits 1:0 **USART1SW[1:0]**: USART1 clock source selection
 This bit is set and cleared by software to select the USART1 clock source.
 00: PCLK selected as USART1 clock source (default)
 01: System clock (SYSCLK) selected as USART1 clock
 10: LSE clock selected as USART1 clock
 11: HSI clock selected as USART1 clock

9.4.14 RCC register map

Table 33. RCC register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	RCC_CR	Res.	Res.	Res.	Res.	Res.	Res.	PLLRDY	PLLON	Res.	Res.	Res.	Res.	CSSON	HSEBYP	HSERD	HSEON	HSICAL[7:0]							HSITRIM[4:0]				Res.	HSIRDY	HSION			
	Reset value							0	0					0	0	0	0	x	x	x	x	x	x	x	x	1	0	0	0	0		1	1	
0x04	RCC_CFGR	PLLNODIV ⁽¹⁾	MCOPRE[2:1] ⁽¹⁾			MCOPRE0 ⁽¹⁾ / MCOF ⁽²⁾	Res.	MCO [2:0]			I2SSRC	USBPRE	PLLMUL[3:0]			PLLXTPRE	PLLSRC	PLLSRC ⁽³⁾	Res.	PPRE2 [2:0]			PPRE1 [2:0]		HPRE[3:0]			SWS [1:0]		SW [1:0]				
	Reset value	0	0	0	0		0	0	0	0	CSSC	0	Res.	0	0	0	0	0	0		0	0	0	0	0		0	0	0	0	0	0	0	
0x08	RCC_CIR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CSSC	0	Res.	Res.	Res.	0	LSIRDYC	Res.	Res.	Res.	PLLRDYC	HSERDYC	HSIRDYC	LSERDYC	LSIRDYC	CSSF	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value										0			0	0	0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	
0x0C	RCC_APB2RSTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIM16RST	TIM15RST	SPI4RST ⁽³⁾	USART1RST	TIM8RST ⁽²⁾	SPI1RST	TIM1RST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYSCFGSRST	
	Reset value															0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x010	RCC_APB1RSTR	Res.	I2C3RST	DAC1RST	PWRRST	Res.	DAC2RST ⁽¹⁾	CANRST	Res.	USBRST ⁽²⁾	I2C2RST ⁽²⁾	I2C1RST	UART5RST ⁽²⁾	UART4RST ⁽²⁾	USART3RST	USART2RST	Res.	SPI3RST	SPI2RST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value			0	0		0	0		0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x14	RCC_AHBENR	Res.	Res.	ADC34EN ⁽²⁾	ADC12EN	Res.	Res.	Res.	TSCEN	Res.	GPIODEN ⁽³⁾	GPIODEN	GPIODEN	GPIODEN	GPIODEN	GPIODEN	GPIODEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value			0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x18	RCC_APB2ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value												0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1C	RCC_APB1ENR	Res.	I2C3EN ⁽³⁾	DAC1EN	PWREN	Res.	DAC2EN ⁽¹⁾	CANEN	Res.	USBEN ⁽²⁾	I2C2EN ⁽²⁾	I2C1EN	UART5EN ⁽²⁾	UART4EN ⁽²⁾	USART3EN	USART2EN	Res.	SP3EN ⁽²⁾	SPI2EN ⁽²⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value			0	0		0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 33. RCC register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x20	RCC_BDCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RTS[1:0]	RTS[1:0]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	
0x24	RCC_CSR	LPWRSTF	WWDGRSTF	IWDGRSTF	SFTRSTF	PORRSTF	PINRSTF	OBLRSTF	RMVF	V18PWRRSTF	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x28	RCC_AHBRSTR	Res.	Res.	ADC34RST ⁽²⁾	ADC12RST	Res.	Res.	Res.	Res.	GPIORST ⁽³⁾	GPIORST	GPIORST ⁽²⁾	GPIODRST	GPIOCRST	GPIOBRST	GPIOARST	GPIOHRST ⁽³⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FMCIRST ⁽³⁾	Res.	Res.	Res.	Res.	Res.
	Reset value			0	0					0	0	0	0	0	0	0	0										0						
0x2C	RCC_CFGR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value																																
0x30	RCC_CFGR3	Res.	Res.	Res.	Res.	Res.	Res.	TIM34SW ⁽³⁾	TIM2SW ⁽³⁾	UART5SW[1:0] ⁽²⁾	UART4SW[1:0] ⁽²⁾	USART3SW[1:0]	USART2SW[1:0]	TIM20SW ⁽³⁾	Res.	Res. ⁽¹⁾	Res.	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	Res. ⁽¹⁾	
	Reset value									0	0	0	0											0	0			0	0			0	0

1. On STM32F303xB/C and STM32F358xC devices only.
2. On STM32F303x6/8 and STM32F328x8 devices only.
3. On STM32F303xD/E only

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.